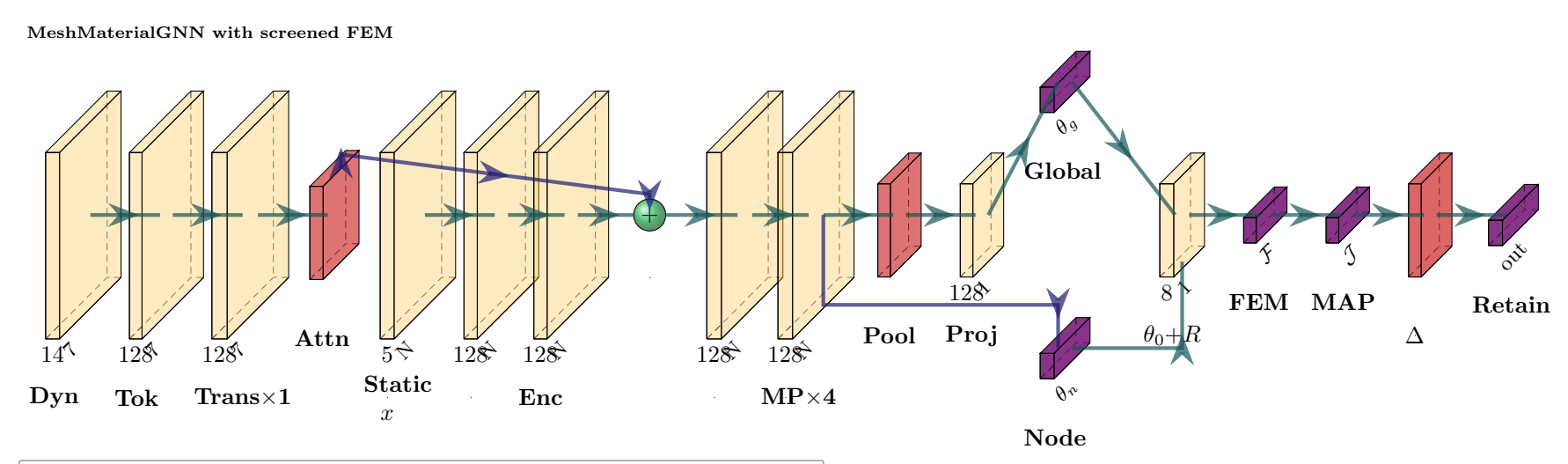

MeshMaterialGNN with screened FEM
14
128
128
Dyn
Tok
Trans×1
Attn
5
128
128
Static
x
Enc
128
128
MP×4
Pool
Proj
128
Global
θ_g
θ_r
Node
8
θ_0+R
$\mathcal{F}$
$\mathcal{J}$
FEM
MAP
Δ
out
Retain
Legend
Feature/MLP (incl. Enc, MP plates)
Pool/Attn/Screen
Heads/FEM/Decision
residual fuse
Flow: Static‖Dyn ⊕ MP×4; Pool→Global; MP→Node (parallel); $\theta_0+R \to \mathcal{F} \to \mathcal{J} \to \Delta \to$Retain.
Blocks: Enc = Lin+SiLU+LN; Attn = view/load; Pool = mean‖max.
Outputs: Global log E_b, log ρ, $\mathbf{c}$, r; Node log E_i, s_i, partition.